Project: To-Do List API **CR ID:** CR-005 **Title:** AI-Assisted Task Breakdown **Type:** Enhancement — Productivity / AI Integration
Priority: Low **Status:** Proposed

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1. Problem Statement

Users may create large, broad tasks (e.g. "Develop Employee Attendance System") without breaking them into smaller, manageable activities. The system currently offers no assistance in identifying whether a task is too large or in suggesting how it could be split up.

2. Proposed Change

Provide an optional, AI-assisted analysis of a task's title, description, and due date that suggests whether the task should be broken down, and if so, proposes a set of smaller candidate tasks (each with its own suggested title, description, and due date). The AI only *suggests* — it does not create any tasks itself. A human (via the frontend) reviews, edits, selects, or discards suggestions, and submits each approved one individually through the existing task-creation endpoint.

3. Scenario

1. A user is about to create a task: "Develop Employee Attendance System," due in about two months, with a description mentioning several distinct pieces of work (registration, fingerprint integration, reporting, admin dashboard).
2. Before submitting it as one task, the user requests an AI breakdown analysis for this title/description/due date.
3. The system calls the configured AI provider and returns an assessment: this task should be broken down, along with several suggested smaller tasks, each with its own title, description, and suggested due date working backward from the original deadline.
4. The frontend displays the suggestions. The user edits one suggested due date, removes one suggestion they don't want, and approves the rest.
5. The frontend submits each approved suggestion as a separate, ordinary `POST /api/tasks` request. The AI breakdown step itself created nothing in the database — only these individual, explicit creation calls did.

4. Functional Requirements

- **FR-01:** The system shall provide an endpoint that accepts a task's title, description, and due date, and returns an AI-generated breakdown assessment.
- **FR-02:** The assessment shall indicate whether a breakdown is recommended (`shouldBreakDown: true/false`) and, if true, a list of suggested subtasks, each with a title, description, and suggested due date.
- **FR-03:** The assessment shall include a brief reason explaining the recommendation.
- **FR-04:** The AI analysis shall **not** persist any task to the database — it returns suggestions only; nothing is created unless the client separately calls the existing task-creation endpoint for each approved suggestion.
- **FR-05:** No new database entities (e.g. parent/subtask relationships) shall be introduced — suggested tasks are plain, independent tasks if and when they are created.
- **FR-06:** If the configured AI provider is unavailable, times out, or returns an invalid response, the breakdown endpoint shall fail gracefully with a clear error, and ordinary task creation (`POST /api/tasks`) shall remain fully operational and unaffected.
- **FR-07:** AI-suggested due dates shall still be subject to the system's existing due-date validation (cannot be in the past) if and when they are submitted as an actual task.
- **FR-08:** The AI provider and credentials shall be externally configurable rather than hard-coded, and credentials shall not be exposed in API responses or logs.

5. Acceptance Criteria

- Submitting a task description that clearly contains multiple distinct activities returns `shouldBreakDown: true` with a non-empty, reasonable list of suggestions.
- Submitting a small, atomic task returns `shouldBreakDown: false` with an empty or minimal suggestion list.
- No row is added to the `TASK` table as a direct result of calling the breakdown endpoint, under any response outcome.
- When the AI provider is simulated as unreachable, the breakdown endpoint returns a clear error and `POST /api/tasks` continues to work normally in the same test run.

- An approved suggestion, submitted through the existing task-creation endpoint with a due date in the past (e.g. due to a stale suggestion), is rejected the same way any other invalid task creation is rejected today.

6. Scope

In Scope:

- A dedicated endpoint that returns AI-generated suggestions for a given task's title/description/due date.
- Configuration for an external AI provider (e.g. DeepSeek).
- Graceful failure handling so AI unavailability never blocks ordinary task creation.

Out of Scope:

- Any database relationship between a task and the suggestions that produced it (no `ParentTask` / `Subtask` entities, per explicit architecture direction).
- Automatic creation of suggested tasks — creation always goes through the existing, unmodified task-creation endpoint, one task at a time, initiated by the client.
- Any specific frontend implementation (out of scope for this backend-focused CR; the frontend's responsibilities are described only to establish the API contract's boundaries).

7. Approval

- **Business Approval:** Pending
- **Technical Approval:** Pending
- **Implementation Status:** Not Started